

Spencer Wallace

☎ 520 461 4480 | @ spencerw530@gmail.com | GitHub | LinkedIn 📍 San Diego, CA

EDUCATION

University of Washington

PhD Astrophysics

Seattle, Washington

August 2023

University of Arizona

BS Computer Science, Astronomy and Physics

Tucson, Arizona

May 2014

EXPERIENCE

Postdoctoral Researcher

University of Washington, Astronomy Department (Remote)

September 2023 – Present

- Developed and optimized a large-scale scientific computing code ([ChaNGa](#)) for heterogeneous CPU-GPU systems, improving scaling on NVIDIA Grace Hopper hardware to over 500+ GPUs and 30,000+ cores
- Profiled and debugged GPU memory and performance bottlenecks with NVIDIA tools, reducing stalls and improving throughput by a factor of 5x
- Collaborated with HPC specialists across physics, CS, and biology to adapt algorithms to diverse workloads

Research Assistant

University of Washington, Astronomy Department

October 2019 – August 2023

- Designed and implemented a parallel tree-based collision detection module in C++/MPI, enabling the first planet formation simulations with realistic-sized bodies, which resulted in three first-author publications
- Contributed to [paratreed](#), a toolkit for testing/tuning distributed tree traversal algorithms, which resulted in a coauthored [IEEE publication](#)
- Developed a data generation pipeline using PyTorch ML models, cutting computation cost of my thesis work by 900k+ CPU hours
- Led and mentored 5 astrophysics students on software development practices, debugging, and data analysis

Teaching Assistant

University of Washington, Astronomy Department

October 2015 – September 2019

- Led weekly discussion sections, graded exams, designed homework assignments and performed guest lectures for undergraduate astronomy classes, some of which involved teaching basic data analysis and numerical modeling skills using Python and Excel
- Collaborated with a team of other teaching assistants to ensure exams and assignments were graded consistently and fairly

Software Consultant

LSST Corporation

June 2011 – August 2014

- Designed full-stack web components (JavaScript/Node.js) for [ASCOT](#), an interactive astronomical visualization platform
- Built a fully customizable, persistent dashboard interface, along with documentation to facilitate large survey data exploration

SKILLS

Programming Languages: C++, Python, Bash, JavaScript

Tools and Environments: Git, CUDA, NumPy, pandas, PyTorch, SLURM, matplotlib, seaborn, node.js

Communication: 3 first-authored publications, 2 co-authored publications, 8 conference talks, 3 conference posters, 20 pop-sci articles published through [astrobites](#)